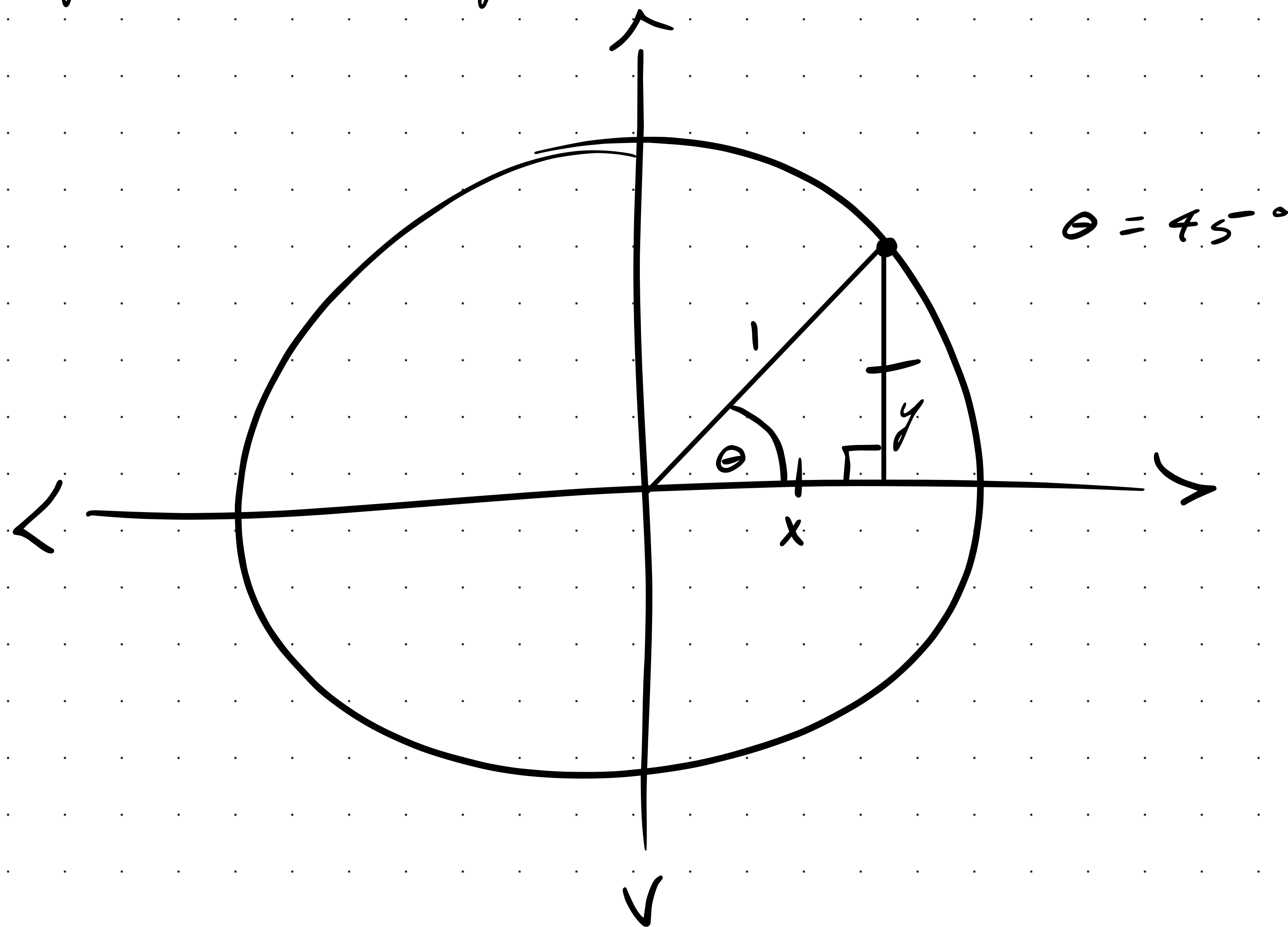


Key Assumptions:

- A triangle's interior angles must add up to 180°



45-45-90

This triangle divides a quadrant in two equal parts since $90/2 = 45$. This means that the triangle's two legs are of equal length since this forms a perfect square.

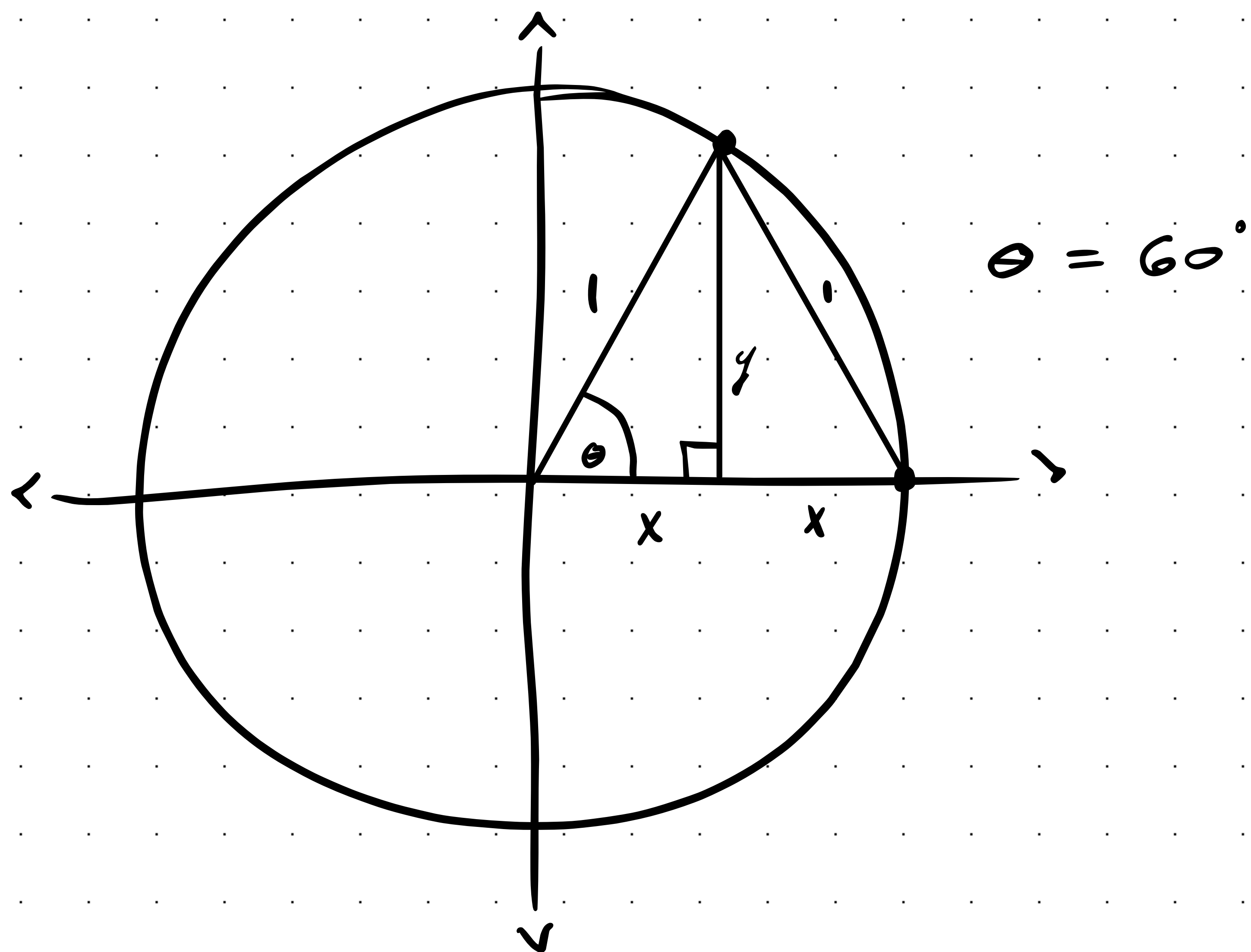
The hypotenuse is of length 1

So $x^2 + y^2 = 1 \rightarrow$ Pythagorean identity

We also know that $x = y$, so $x^2 + x^2 = 1 \rightarrow 2x^2 = 1 \rightarrow x = \sqrt{\frac{1}{2}}$

$$\rightarrow x = \frac{\sqrt{1}}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\therefore x, y = \frac{\sqrt{2}}{2}$$



30-60-90

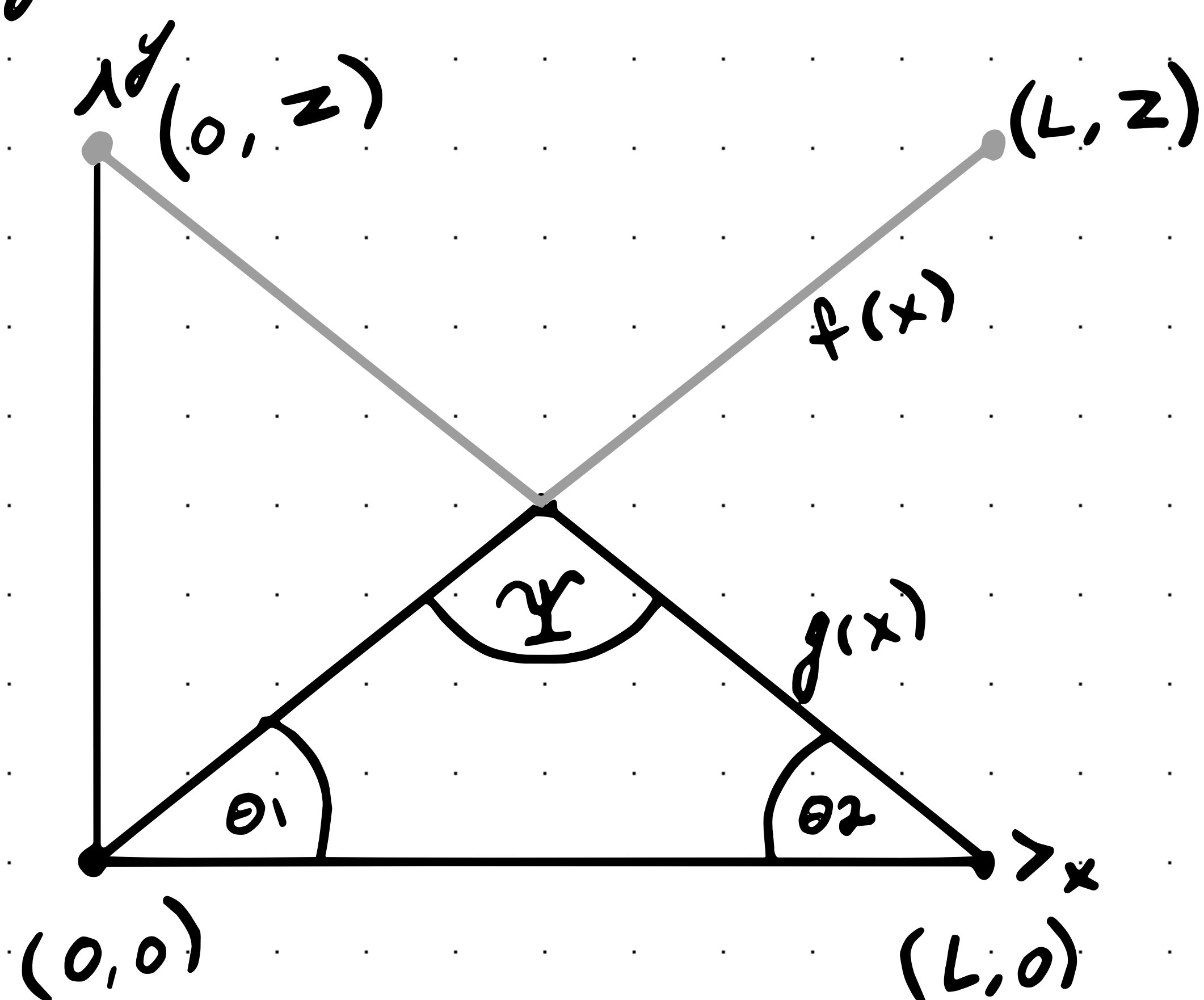
Reflecting the original triangle across its y -leg produces a triangle with three 60° angles. θ is 60° , the top is 60° because $30^\circ \cdot 2 = 60^\circ$, and the last angle is a reflection of θ . These three add to 180° .

We know that two of the triangle's sides are one unit in length

Here we will prove that, for any triangle, if any two of its angles are equal, then so too must be the two sides opposite these angles.

Proof

We will anchor an arbitrary triangle with two equal angles, θ_1 and θ_2 , to a coordinate plane as follows:



Let
 L = triangle's base length

This being done, we can extend the triangle's non-base sides to arbitrary distances, and we can call the resulting lines $f(x)$ and $g(x)$.

$f(x)$ and $g(x)$ are linear functions of equal slope. This is necessarily the case since $\theta_1 = \theta_2$, which is to say that the angle formed by $f(x)$ and $g(x)$ with the x -axis are equal, and so too are the slopes.

We will extend $g(x)$, the negatively-sloped line, until it intersects the y -axis at the point $(0, z)$. Extend $f(x)$ to the point (L, z) .

We can represent $f(x)$ and $g(x)$ between $x=0$ and $x=L$ (the restricted domain of interest) as follows:

$$f(x) = \frac{z}{L} x$$

$$g(x) = z - \frac{z}{L} x$$

Since the slope between these points is $\frac{z}{L}$ and $-\frac{z}{L}$ respectively.

Now we will find the point at which these lines intersect using algebra.

$$f(x) = g(x) \rightarrow \frac{z}{L} x = z - \frac{z}{L} x \rightarrow 2 \frac{z}{L} x = z$$

$$\rightarrow x = \frac{Lz}{2z} = \frac{L}{2}$$

◦ The point at which $f(x)$ and $g(x)$ intersect is

at $x = \frac{L}{2}$, or halfway along the length of the triangle's base.

Inspecting the lines on either side of $x = \frac{L}{2}$, we have:

$$f\left(\frac{L}{2}\right) - f(0) = \frac{z}{L}\left(\frac{L}{2}\right) - \frac{z}{L}(0) = \frac{zL}{2L} = \frac{z}{2}$$

$$\begin{aligned} g\left(\frac{L}{2}\right) - g(L) &= \left(z - \frac{z}{L}\left(\frac{L}{2}\right)\right) - \left(z - \frac{z}{L}(L)\right) \\ &= \left(z - \frac{zL}{2L}\right) - (0) = z - \frac{z}{2} = \frac{z}{2} \end{aligned}$$

$$\frac{z}{2} = \frac{z}{2}$$

∴ The height of each side is equal.

We now know that the length of each side is $\frac{L}{2}$ and the height of each side is $\frac{z}{2}$.

We know that we can split the triangle at $x = \frac{L}{2}$ with a line perpendicular to the x-axis and produce two equal right-triangles since we have just showed that the shapes on either side of this line are equal, and we know we have a point at $x = \frac{L}{2}$ by the triangles' definition.

call the shared height of these side lengths h and the length r . Then

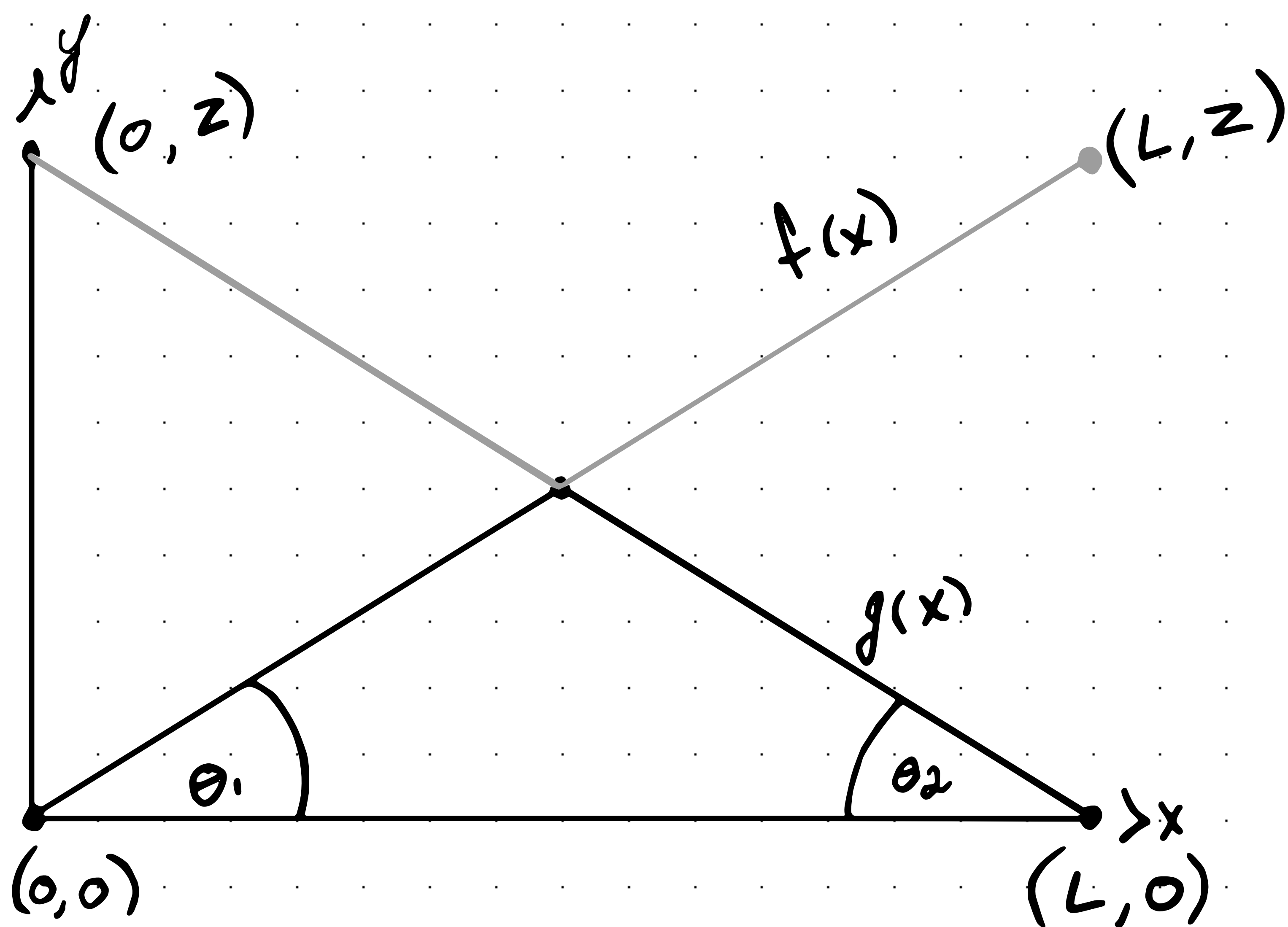
$$r^2 = \left(\frac{L}{2}\right)^2 + h^2 \quad \text{by Pythagorean theorem}$$

$$\rightarrow \ell = \sqrt{\left(\frac{c}{2}\right)^2 + a^2} =$$

We have shown that, given any triangle with two equal angles, the side lengths opposite these angles must also be equal and have a length of

$$\ell = \sqrt{\left(\frac{c}{2}\right)^2 + a^2}$$

Q.E.D.



z : y -value at which $g(x)$ intersects the y -axis; arbitrary

L : x -value at which $f(x)$ intersects the x -axis; arbitrary

$$f(x) = \frac{z}{L}x, \quad g(x) = z - \frac{z}{L}x$$

$$\frac{z}{L}x = z - \frac{z}{L}x$$

$$2\frac{z}{L}x = z$$

$$x = \frac{z}{\frac{2z}{L}} = \frac{Lz}{2z} = \frac{L}{2}$$

How does showing $f(x) = g(x)$ at $x = \frac{L}{2}$ prove that side one is of equal length to side 2?

Same slope = same hypotenuse $\rightarrow$ same $\frac{1}{2}$ hypotenuse, done